

DOCTOR PLANTS

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Abstract—The rise of smart agriculture has created new opportunities to enhance crop health monitoring through the integration of artificial intelligence (AI) and Internet of Things (IoT). This project, titled "DR Plants", introduces a real-time, AI-powered system for detecting tomato leaf diseases using the ESP32-CAM module. The system utilizes a Convolutional Neural Network (CNN) model trained in TensorFlow to identify diseases such as early blight, late blight, and leaf spot from captured leaf images. The model is deployed on an ESP32-CAM board, allowing image acquisition and disease prediction directly from the device. The detection results are instantly transmitted to the user via the Blynk mobile application, while a push button allows manual triggering of image capture. By combining hardware and software in a compact and low-cost setup, this solution empowers farmers with accessible tools for early disease detection, reducing the need for expert inspection and improving crop management efficiency. The proposed system supports the development of smart farming technologies that are scalable, accurate, and easy to implement

Keywords — *Artificial Intelligence (AI), Tomato Leaf Disease, ESP32-CAM, TensorFlow, Blynk, Smart Agriculture, Image Classification, Deep Learning, CNN, IoT*

I. INTRODUCTION

In recent years, the coupling of AI techniques with embedded systems has broken new ground in precision farming. Early detection and diagnosis of plant diseases is one of the problems in modern farming with potential large scale crop loss if undiagnosed for a long time. As one of the highest valued vegetables in commercial and home gardens, tomatoes are susceptible to a number of diseases, such as early blight, late blight, and leaf spots, which have a direct effect on yield and quality [1].

DR Plants is a project that introduces an unique AI-based solution to tomato leaf disease detection through an ESP32-CAM module. This intelligent system relies on the TensorFlow [3] library to process the images taken with ESP32-CAM through a pre-trained CNN model. The model, developed and trained using PyCharm, is capable of classifying different types of tomato diseases based on visible symptoms on the leaves [1][2]. Upon detection, the result is sent to the Blynk mobile application [4], enabling real-time monitoring for farmers or greenhouse operators. A physical push button is also included to initiate the image capture and detection process manually.

This system offers a low-cost, real-time, and portable solution that empowers farmers with timely insights, reducing dependency on expert knowledge and enabling quicker response to disease outbreaks [2]. The use of cloud-based AI combined with IoT enhances agricultural productivity and supports sustainable farming practices.

PROBLEM STATEMENT

Tomato plants are highly susceptible to various leaf diseases such as early blight, late blight, and leaf spot, which can significantly reduce crop yield and quality if not detected and treated promptly. In traditional farming, disease diagnosis relies heavily on manual observation and expert knowledge, which may not always be available, especially in rural or underdeveloped areas. This manual approach is often time-consuming, inconsistent, and prone to human error.

Additionally, small-scale farmers may lack access to advanced diagnostic tools or laboratories, limiting their ability to identify diseases accurately and early. While AI-based plant disease detection solutions exist, many are expensive, require internet

connectivity at all times, or demand complex hardware setups not suitable for field deployment [3].

Therefore, there is a critical need for a low-cost, portable, and real-time disease detection system that is easy to use, does not depend on continuous internet access, and provides fast results directly to the farmer. This project, "DR Plants", addresses these challenges by integrating a deep learning model with an ESP32-CAM and mobile application, providing an accessible and efficient solution for disease monitoring in tomato cultivation.

OBJECTIVE

1. To develop an AI-based system using TensorFlow for detecting and classifying tomato leaf diseases through image processing.
2. To implement the trained deep learning model on an ESP32-CAM module for real-time image capture and disease diagnosis.
3. To integrate the detection system with the Blynk mobile application for user-friendly and remote monitoring of plant health.

II. METHODOLOGY

A. Software



Figure 1: TensorFlow

TensorFlow is a machine learning framework that has offered flexibility, scalability and performance for deep learning tasks. tf. function helps to optimize and accelerate computation by leveraging graph-based execution.[4]



Figure 2: PyCharm

PyCharm is an IDE (Integrated Development Environment) specifically designed for Python development, offering a wide range of tools to enhance coding, debugging, and testing.



Figure 3: Blynk

Blynk is a low-code Internet of Things (IoT) platform that allows users to quickly prototype, deploy, and manage connected devices.[5]

B. Hardware



Figure 4: ESP32-CAM

The ESP32-CAM is a compact, low-cost development board with built-in Wi-Fi and Bluetooth, designed for various Internet of

Things (IoT) applications, particularly those involving visual data.



Figure 5: LED

Light Emitting Diode is a semiconductor based light source that emits light whenever current flows through it.



Figure 6: Push Button

A push button's primary function is to control the flow of electricity by momentarily connecting or disconnecting two or more contacts.

C. Algorithm

Start

1. Initialize:

- Serial Communication
- WiFi.
- ESP32-CAM.
- AI Model.
- LED.
- Blynk Application

2. Loop:

If Push Button is Pressed:

- Turn ON Green LED

- Capture image using ESP32-CAM

- Send image to AI server for analysis

- Receive classification results on Blynk or website.

If; plant is healthy:

- Display "Healthy" on Blynk app

- Suggest routine care tips.

Else; plant is diseased

- Display disease name and care suggestion on Blynk app

Return to loop

End

D. Flowchart

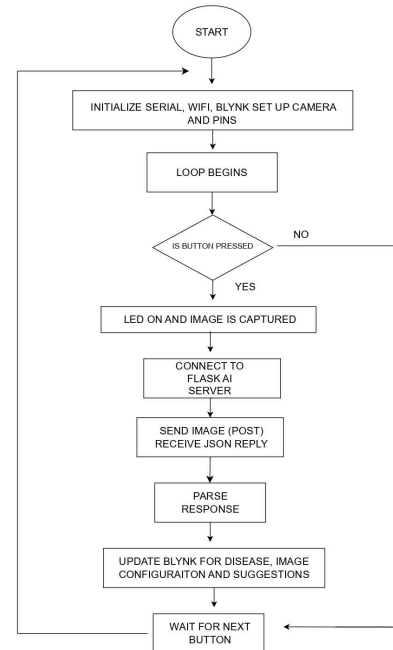


Figure 7: Flowchart of Doctor Plants

The AI system flowchart shown in Figure 7 illustrates the operation of the leaf disease detection process.

The system starts by initializing all components, including the ESP32-CAM, WiFi, and Blynk connection. Then, when the push button is pushed, a green LED will turn on to indicate that image has been captured and sent to the AI server for disease analysis. Based on the diagnosis result, suggestion actions: if the plant is healthy, a message is sent to the Blynk app; if a disease is detected, the diagnosis with a care suggestion is displayed in the app. The system then will return to standby mode, and the process is repeated until the push button is pressed again.

E. Block Diagram

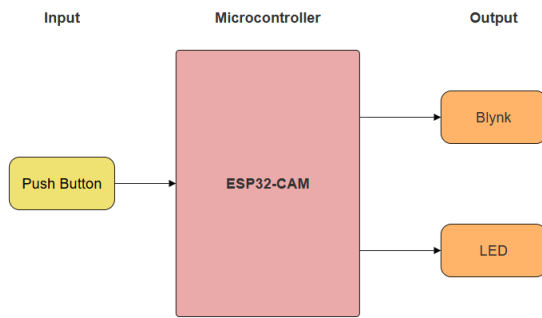


Figure 8: Block Diagram of Doctor Plant

The block diagram explains the functional flow of our project tomato disease detection system using an ESP32-CAM as a microcontroller. The system is divided into three main parts, the first part is Input, second part is Microcontroller and the last part is Output.

The input part is a push button. This button serves as a manual trigger for the system. When pressed, it gives the signal to the ESP32-CAM to capture an image of the tomato leaf whether it has disease or not. Basically it is called disease analysis. This allows the user to control when the detection should be performed, rather than using continuous or automated image capture.

The second part is a microcontroller, ESP32-CAM is the core processing unit of the system. It performs the system. It also performs multitasking. For example, you can capture the image using its built-in camera, run a pre-trained AI model to analyze the image and detect any visible disease on the leaf [1] [2]. Not only that ESP32-CAM also sends the analysis result to

external output such as the mobile app or indicators for this project we use Blynk

The last is output, Blynk is output for this project. The flow is once the image is analyzed, the detection result is sent to the Blynk application over Wi-Fi for example early blight, late blight, curl virus. This allows users to receive real-time notifications. Next, LED serves as a visual indicator of the result. This project, we use green LED as indicators for successfully capturing the image by ESP32-CAM [11].

In short, the push button activates the ESP32-CAM, which processes the image and sends the result to both Blynk and LED. This makes your system user-friendly, portable, and ideal for smart agriculture applications in remote areas.

F. Schematic Diagram

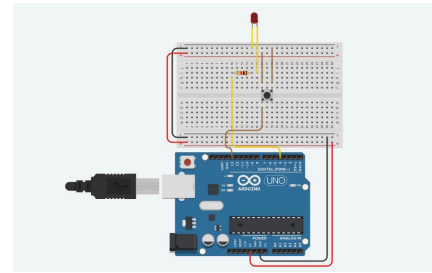


Figure 9

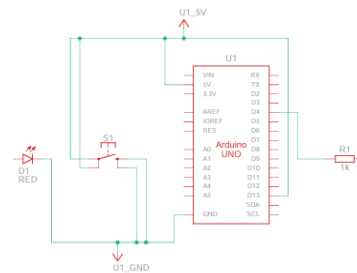


Figure 10

In figure 10 the component we use is the Arduino UNO as the ESP32-CAM because in the software there is no ESP32-CAM so we decide to use that component. S1 as the push button. R1 as the resistor and D1 RED as the green LED. The LED is connected to GPIO 4 with a current limiting resistor while the push button is connected to GPIO 13 and

ground. When the push button sends a signal to GPIO 13, ESP32-CAM detects the push button and turns ON the LED. When the button is released, the LED turns OFF.

G. Gantt Chart

TASK	WEEK													
	6	7	8	9	10	11	12	13	14					
Plant image collection and folder organization														
AI model training using Pycharm														
Connect and train model using TensorFlow														
Set up ESP32-CAM and test the AI model														
Connect AI to Blynk for image testing.														
Continue coding for plant care recommendation.														
Test Blynk application														
Integrate led														
Organize coding and fix bugs														
Test the whole system														
Start on prototype														
Final testing and improvement														
Prepare report and presentation														

AI model will analyze and classify whether the tomato leaf is healthy or not and send the results via HTTPS to a flask based server hosted online. The AI model also will provide confidence score and care recommendations [9].



Figure 11: Prototype

III. EXPECTED RESULT AND DISCUSSION

This project is an AI based plant monitoring system that focuses on image processing, internet of things and application integration as well as web platforms to ensure it is user friendly [7]. The main goal is to detect and diagnose tomato disease which are late blight, early blight, curl virus and bacterial spot based on their leaf using the trained ai model. The Convolutional Neutral Network (CNN) model was chosen for the AI due to its strong image classification and was trained using TensorFlow which was then implemented in a low cost ESP32-cam module [8].

To make it easier to operate, the prototype is equipped with a push button to capture the image as shown in figure 11. When the image is captured or the push button is pressed, the green LED will turn on to indicate the image has been captured. The

Then, this information is returned in JSON format to Blynk application via ESP32-cam through virtual pin. Moreover, the server provides the attempt history, which can be downloaded in CSV format. The downloaded csv includes the date and time of each detection, disease category results, confidence percentages, care suggestions, and links [10].

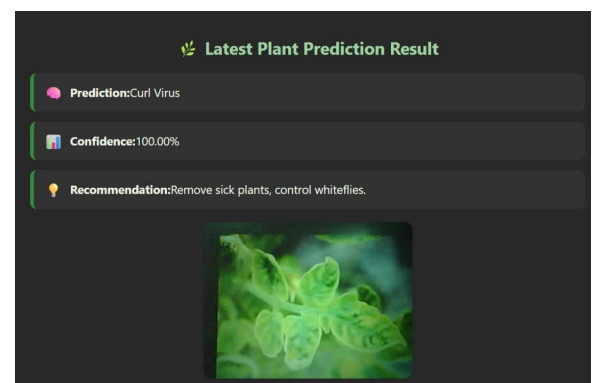


Figure 12: Results that come out from the website.

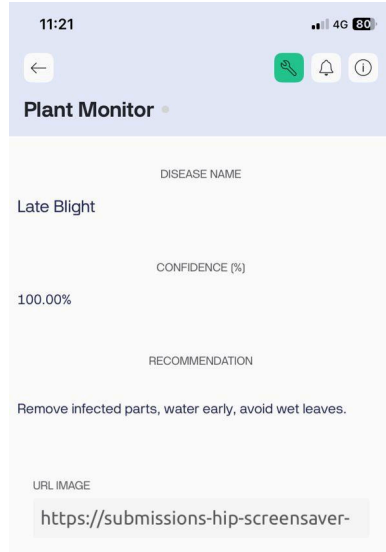


Figure 13: Results that appear in Blynk App.

From figure 12 and figure 13, the output is the same. For the website, we can see that picture directly from there, meanwhile for the Blynk we need to tap the link and a picture will appear. Another reason why we make two platforms for the result is because Blynk App for the phone has easy ways for users to monitor from via the phone without not opening the laptop.

Next, in our Blynk app or the websites, there are names of the diseases like early blight, late blight, curl virus based on what diseases on that leaf. Confidence in percentage, more the percentage value of those diseases so that means the diseases are very strong, for example on the figure 11 the confidence shows 100.00% so it means we sure the leave has diseases named late blight. After that recommendation, our Blynk app or website will suggest what to do when tomato leaves have diseases. The suggestion is based on the diseases on that plant so it is easy to use to take care of their tomato plants.

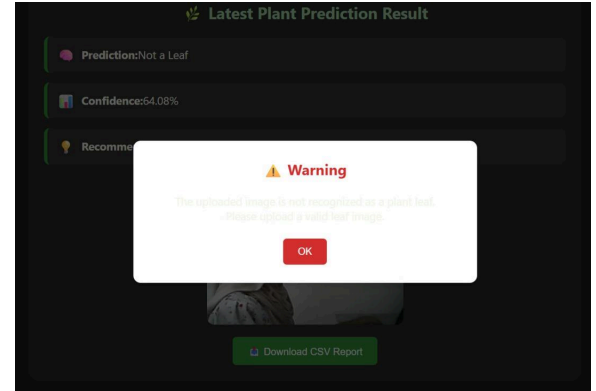


Figure 114: When ESP32-CAM doesn't detect leaf

Basically, if the user snaps other than leaves. Our website will give the warning “The uploaded image is not recognized as a plant leaves. Please upload a valid leaf image”. Our project will detect the leaves of tomato only other than that it cannot detect.

Table 1: Table of dark and light mode

Figure 15:Dark Mode	Figure 16:Light Mode

In table one shows a different color if the user clicks the button for “dark mode”, the website will change to figure 15 while user clicks back on the “dark mode” button it will change to figure 16. This website is also friendly for users that are sensitive to light. Next, users also can download the picture that ESP32-CAM captures in the “Prediction History”. Users just search what time or date they want to download. It is easy for users to handle this website.

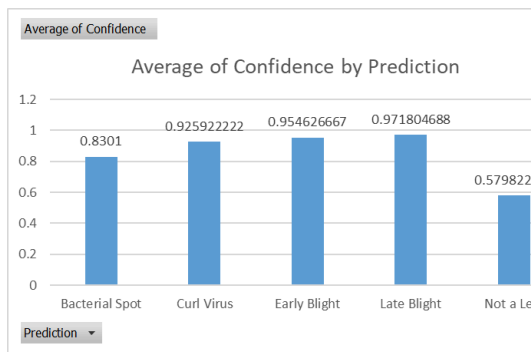


Figure 17: Average confidence of AI disease detection on tomato leaves.

Figure 17 above is a bar chart used to display average confidence level for each disease detected by the AI model. From the results, late blight and early blight were among the highest average confidence with 0.97 and 0.95 each. This could be due to the sharp or clear visual and feature in the images dataset used to train the model. Curl virus also indicates a strong accuracy based on the high confidence of 0.93. Bacterial spots on the other hand recorded a lower average confidence, 0.83 which may indicate that the leaf dataset is visually similar or unclear compared to other disease's dataset. Lastly the lowest confidence 0.58 is recorded by "not a leaf". This could be due to the model having difficulty to differentiate non leaf objects because of background noise or the absence of non leaf examples in the dataset. In addition, these results will allow us to do analysis of trends over time to improve the model in the future.

IV. CONCLUSION

In conclusion we have successfully developed an AI-based plant detection system using TensorFlow and the ESP32-CAM to detect and classify tomato leaf diseases. By integrating the trained deep learning model to Blynk and Flask server, accurate diagnosis were obtained and displayed to users via a user friendly

application so that further action can be taken immediately. In addition the LED and web-based image with downloadable csv report for history access were added to enhance interaction as well as allowing further analysis on user end. This project has demonstrated the integration of AI and IoT that is beneficial in the agriculture industry. Although the system requires a stable internet connection, the overall prototype demonstrated strong potential as a low-cost and wireless solution for early disease detection. With more improvement, this system can be potentially optimized for agricultural use in real life.

V. RECOMMENDATIONS

To improve the Dr Plant project, we can improve the network connectivity by exploring offline functionality as the current system depends on a strong and stable WiFi connection. In agricultural plantation areas where the internet is unavailable, this issue can be the major limitation. A few AI trainers that would allow the model to be deployed directly on ESP32-CAM so that offline processing is possible such as Edge Impulse can be considered. Next, to improve accuracy of detected disease, the dataset used should be more diverse and balanced. Adding the negative sample for non leaf will also make the model able to distinguish the leaf better and enhance the overall performance of the AI.

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